Partial Fractions and Improper Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Improper integrals: infinite limits**

Evaluate $\int_1^\infty \frac{1}{x^2}\,dx$, or show it diverges.

$\int_1^\infty \frac{1}{x^2}\,dx = \lim_{b\to\infty}\left[-\frac{1}{x}\right]_1^b = \lim_{b\to\infty}\left(-\frac{1}{b}+1\right) = 0+1 = 1$. Converges to 1.

2 **[Foundational]** • **Target: Visualizing convergence: area under the curve**

By contrast, $\int_1^\infty \frac{1}{x}\,dx$ diverges. Compare the graphs of $y=\frac{1}{x}$ and $y=\frac{1}{x^2}$ on $[1,10]$ and explain why $\frac{1}{x}$ decays too slowly for the area to stay finite.

Both curves decrease, but $\frac{1}{x}$ stays larger than $\frac{1}{x^2}$ for all $x>1$ (e.g. at $x=10$: 0.1 vs 0.01). This slower decay means the accumulated area under $\frac{1}{x}$ keeps growing without bound as $x\to\infty$, while $\frac{1}{x^2}$'s faster decay allows its total area to converge.

3 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Decompose $\frac{5x-1}{(x-1)(x+2)}$ into partial fractions.

$\frac{5x-1}{(x-1)(x+2)} = \frac{A}{x-1}+\frac{B}{x+2}$. $5x-1 = A(x+2)+B(x-1)$. At $x=1$: $4=3A \Rightarrow A=\frac{4}{3}$. At $x=-2$: $-11=-3B \Rightarrow B=\frac{11}{3}$. So $\frac{4/3}{x-1}+\frac{11/3}{x+2}$.

4 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Evaluate $\int \frac{3x+5}{x^2+x-2}\,dx$ using partial fractions (note $x^2+x-2=(x+2)(x-1)$).

$\frac{3x+5}{(x+2)(x-1)} = \frac{A}{x+2}+\frac{B}{x-1}$. $3x+5 = A(x-1)+B(x+2)$. At $x=1$: $8=3B \Rightarrow B=\frac{8}{3}$. At $x=-2$: $-1=-3A \Rightarrow A=\frac{1}{3}$. $\int\left(\frac{1/3}{x+2}+\frac{8/3}{x-1}\right)dx = \frac{1}{3}\ln|x+2|+\frac{8}{3}\ln|x-1|+C$.

5 **[Foundational]** • **Target: Partial fraction decomposition: distinct linear factors**

Decompose $\frac{x+7}{x^2-x-6}$ into partial fractions (note $x^2-x-6=(x-3)(x+2)$).

$\frac{x+7}{(x-3)(x+2)} = \frac{A}{x-3}+\frac{B}{x+2}$. $x+7 = A(x+2)+B(x-3)$. At $x=3$: $10=5A \Rightarrow A=2$. At $x=-2$: $5=-5B \Rightarrow B=-1$. So $\frac{2}{x-3}-\frac{1}{x+2}$.

6 **[Foundational]** • **Target: Partial fractions with a repeated linear factor**

Decompose $\frac{2x+3}{(x-1)^2}$ into partial fractions.

$\frac{2x+3}{(x-1)^2} = \frac{A}{x-1}+\frac{B}{(x-1)^2}$. $2x+3 = A(x-1)+B$. At $x=1$: $5=B$. Comparing x coefficients: $A=2$. So $\frac{2}{x-1}+\frac{5}{(x-1)^2}$.

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7 [Foundational] • Target: Partial fractions with a repeated linear factor

Evaluate $\int \frac{2x+3}{(x-1)^2}\,dx$ using the decomposition found above.

$\int \left(\frac{2}{x-1} + \frac{5}{(x-1)^2}\right)dx = 2\ln|x-1| - \frac{5}{x-1} + C.$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Partial fractions with an irreducible quadratic factor

Set up (but do not solve for constants) the partial fraction decomposition of $\frac{3x^2+2}{(x-1)(x^2+4)}$.

Since x^2+4 is irreducible over the reals, the decomposition has the form $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$.

9 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_1^\infty \frac{1}{x}\,dx$, or show it diverges.

$\int_1^\infty \frac{1}{x}\,dx = \lim_{b\to\infty}[\ln x]_1^b = \lim_{b\to\infty}(\ln b - 0) = \infty$. Diverges.

10 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_0^\infty e^{-2x}\,dx$, or show it diverges.

$\int_0^\infty e^{-2x}\,dx = \lim_{b\to\infty}\left[-\frac{1}{2}e^{-2x}\right]_0^b = \lim_{b\to\infty}\left(-\frac{1}{2}e^{-2b} + \frac{1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}$. Converges to $\frac{1}{2}$.

11 [Intermediate] • Target: Improper integrals: infinite limits

Evaluate $\int_{-\infty}^0 e^x\,dx$, or show it diverges.

$\int_{-\infty}^0 e^x\,dx = \lim_{a\to-\infty}[e^x]_a^0 = \lim_{a\to-\infty}(1 - e^a) = 1 - 0 = 1$. Converges to 1.

12 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^1 \frac{1}{\sqrt{x}}\,dx$, or show it diverges (note the integrand is undefined at $x = 0$).

$\int_0^1 x^{-1/2}\,dx = \lim_{a\to0^+}[2\sqrt{x}]_a^1 = \lim_{a\to0^+}(2 - 2\sqrt{a}) = 2 - 0 = 2$. Converges to 2.

13 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^1 \frac{1}{x}\,dx$, or show it diverges.

$\int_0^1 \frac{1}{x}\,dx = \lim_{a\to0^+}[\ln x]_a^1 = \lim_{a\to0^+}(0 - \ln a) = \infty$. Diverges.

14 [Intermediate] • Target: Improper integrals: discontinuous integrands

Evaluate $\int_0^2 \frac{1}{(x-2)^2}\,dx$, or show it diverges (the integrand is undefined at $x = 2$).

$\int_0^2 (x-2)^{-2}\,dx = \lim_{b\to2^-}\left[-\frac{1}{x-2}\right]_0^b = \lim_{b\to2^-}\left(-\frac{1}{b-2} - \frac{1}{2}\right) = \infty$. Diverges.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The p-test for improper integrals • Cross-Lesson: 4.02 Area in Polar Coordinates

State the p-test for $\int_1^\infty \frac{1}{x^p}\,dx$, and use it to determine convergence for $p = 3$ and $p = 0.5$.

$\int_1^\infty \frac{1}{x^p}\,dx$ converges if $p > 1$ and diverges if $p \le 1$. For $p = 3 > 1$: converges. For $p = 0.5 \le 1$: diverges.

16 [Challenging] • Target: The p-test for improper integrals • Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests

State the analogous p-test for $\int_0^1 \frac{1}{x^p}\,dx$ (a discontinuity at $x = 0$), and determine convergence for $p = \frac{1}{2}$ and $p = 2$.

$\int_0^1 \frac{1}{x^p}\,dx$ converges if $p < 1$ and diverges if $p \ge 1$ (opposite condition from the infinite-limit case). For $p = \frac{1}{2} < 1$: converges. For $p = 2 \ge 1$: diverges.

17 [Challenging] • Target: Combining partial fractions with improper integrals • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

Evaluate $\int_0^\infty \frac{1}{(x+1)(x+2)}\,dx$ using partial fractions.

$\frac{1}{(x+1)(x+2)} = \frac{1}{x+1} - \frac{1}{x+2}$. $\int_0^b \left(\frac{1}{x+1} - \frac{1}{x+2}\right)dx = \left[\ln\frac{x+1}{x+2}\right]_0^b = \ln\frac{b+1}{b+2} - \ln\frac{1}{2}$. As $b \to \infty$, $\frac{b+1}{b+2} \to 1$, so the limit is $\ln 1 - \ln\frac{1}{2} = \ln 2$. Converges to $\ln 2$.

18 [Challenging] • Target: Three distinct linear factors • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Decompose $\frac{6x^2 - 3x - 1}{x(x-1)(x+1)}$ into partial fractions.

$\frac{6x^2 - 3x - 1}{x(x-1)(x+1)} = \frac{A}{x} + \frac{B}{x-1} + \frac{C}{x+1}$. $6x^2 - 3x - 1 = A(x-1)(x+1) + Bx(x+1) + Cx(x-1)$. At $x = 0$: $-1 = -A \Rightarrow A = 1$. At $x = 1$: $2 = 2B \Rightarrow B = 1$. At $x = -1$: $8 = 2C \Rightarrow C = 4$. So $\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}$.

19 [Challenging] • Target: Three distinct linear factors • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Evaluate $\int \frac{6x^2 - 3x - 1}{x(x-1)(x+1)}\,dx$ using the decomposition above.

$\int \left(\frac{1}{x} + \frac{1}{x-1} + \frac{4}{x+1}\right)dx = \ln|x| + \ln|x-1| + 4\ln|x+1| + C$.

20 [Challenging] • Target: Visualizing convergence: area under the curve • Cross-Lesson: 1.01 Integration by Parts

The convergent improper integral $\int_1^\infty \frac{1}{x^2}\,dx = 1$ (shown earlier) means the infinite region under $y = \frac{1}{x^2}$ for $x \ge 1$ has FINITE area. Explain intuitively why this is possible even though the region extends forever.

Even though the region is infinitely long, the height $\frac{1}{x^2}$ shrinks fast enough as $x \to \infty$ that the accumulated area still converges to a finite total — the 'tail' contributes less and less area, summing to a bounded amount, similar to how a convergent geometric series sums to a finite value despite infinitely many terms.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Partial fraction decomposition: distinct linear factors

For $\frac{7x+5}{(x+3)(x-4)} = \frac{A}{x+3} + \frac{B}{x-4}$, find A.

a. $-\frac{16}{7}$

b. $\frac{23}{7}$

c. $\frac{16}{7}$ ✓

d. $\frac{33}{7}$

Using the cover-up method, set $x = -3$: $A = \frac{7(-3)+5}{-3-(4)} = \frac{-16}{-7} = \frac{16}{7}$.

22 [Foundational] • Target: Partial fractions with a repeated linear factor

For $\frac{7x-5}{(x)(x+5)} = \frac{A}{x} + \frac{B}{x+5}$, find A.

a. 8

b. -1 ✓

c. 1

d. 0

Using the cover-up method, set $x = 0$: $A = \frac{7(0)-5}{0-(-5)} = \frac{-5}{5} = -1$.

23 [Foundational] • Target: Partial fractions with an irreducible quadratic factor

For $\frac{2x-6}{(x)(x+4)} = \frac{A}{x} + \frac{B}{x+4}$, find A.

a. $\frac{7}{2}$

b. $-\frac{3}{2}$ ✓

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

Using the cover-up method, set $x = 0$: $A = \frac{2(0)-6}{0-(-4)} = \frac{-6}{4} = -\frac{3}{2}$.

24 [Foundational] • Target: Improper integrals: infinite limits

For $\frac{7x+1}{(x-3)(x-2)} = \frac{A}{x-3} + \frac{B}{x-2}$, find A.

a. 23

b. -22

c. -15

d. 22 ✓

Using the cover-up method, set $x = 3$: $A = \frac{7(3)+1}{3-(2)} = \frac{22}{1} = 22$.

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25 [Foundational] • Target: Improper integrals: discontinuous integrands

For $\frac{6x+9}{(x+1)(x-5)} = \frac{A}{x+1} + \frac{B}{x-5}$, find A.

a. $\frac{1}{2}$

b. $\frac{13}{2}$

c. $-\frac{1}{2}$ ✓

d. -1

Using the cover-up method, set $x = -1$: $A = \frac{6(-1)+9}{-1-(5)} = \frac{3}{-6} = -\frac{1}{2}$.

26 [Foundational] • Target: The p-test for improper integrals

For $\frac{3x+3}{(x+3)(x-5)} = \frac{A}{x+3} + \frac{B}{x-5}$, find A.

a. $\frac{7}{4}$

b. $-\frac{3}{4}$

c. $\frac{9}{4}$

d. $\frac{3}{4}$ ✓

Using the cover-up method, set $x = -3$: $A = \frac{3(-3)+3}{-3-(5)} = \frac{-6}{-8} = \frac{3}{4}$.

27 [Foundational] • Target: Combining partial fractions with improper integrals

For $\frac{3x-4}{(x-4)(x-5)} = \frac{A}{x-4} + \frac{B}{x-5}$, find A.

a. -8 ✓

b. 8

c. -7

d. 11

Using the cover-up method, set $x = 4$: $A = \frac{3(4)-4}{4-(5)} = \frac{8}{-1} = -8$.

28 [Foundational] • Target: Three distinct linear factors

Evaluate $\int_1^{\infty} \frac{2}{x^3}\,dx$, or show it diverges.

a. -1

b. 1 ✓

c. 2

d. $\frac{2}{3}$

$\int_1^{\infty} 2x^{-3}\,dx = \lim_{b\to\infty}\left[\frac{2x^{-2}}{-2}\right]_1^b = \lim_{b\to\infty}\left(\frac{-2}{(2)b^2} + \frac{2}{2}\right) = 0 + \frac{2}{2} = 1$. Converges.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Visualizing convergence: area under the curve

Evaluate $\int_1^\infty \frac{8}{x^2}\,dx$, or show it diverges.

a. -8

b. 4

c. 8 ✓

d. 9

$\int_1^\infty 8x^{-2}\,dx = \lim_{b\to\infty}\left[\frac{8x^{-1}}{-1}\right]_1^b = \lim_{b\to\infty}\left(\frac{-8}{(1)b^1}+\frac{8}{1}\right) = 0+\frac{8}{1} = 8$. Converges.

30 [Intermediate] • Target: Improper integrals with both an infinite bound and a discontinuity

Evaluate $\int_1^\infty \frac{3}{x^2}\,dx$, or show it diverges.

a. $\frac{3}{2}$

b. -3

c. 4

d. 3 ✓

$\int_1^\infty 3x^{-2}\,dx = \lim_{b\to\infty}\left[\frac{3x^{-1}}{-1}\right]_1^b = \lim_{b\to\infty}\left(\frac{-3}{(1)b^1}+\frac{3}{1}\right) = 0+\frac{3}{1} = 3$. Converges.

31 [Intermediate] • Target: Comparison test for improper integrals

Evaluate $\int_1^\infty \frac{4}{x^2}\,dx$, or show it diverges.

a. 5

b. -4

c. 2

d. 4 ✓

$\int_1^\infty 4x^{-2}\,dx = \lim_{b\to\infty}\left[\frac{4x^{-1}}{-1}\right]_1^b = \lim_{b\to\infty}\left(\frac{-4}{(1)b^1}+\frac{4}{1}\right) = 0+\frac{4}{1} = 4$. Converges.

32 [Intermediate] • Target: Linear Partial Fractions

Evaluate $\int_1^\infty \frac{2}{x^4}\,dx$, or show it diverges.

a. $\frac{2}{3}$ ✓

b. $\frac{5}{3}$

c. $\frac{1}{2}$

d. $-\frac{2}{3}$

$\int_1^\infty 2x^{-4}\,dx = \lim_{b\to\infty}\left[\frac{2x^{-3}}{-3}\right]_1^b = \lim_{b\to\infty}\left(\frac{-2}{(3)b^3}+\frac{2}{3}\right) = 0+\frac{2}{3} = \frac{2}{3}$. Converges.

33 [Intermediate] • Target: Repeated Linear & Irreducible Factors

Evaluate $\int_1^\infty \frac{8}{x^4}\,dx$, or show it diverges.

a. $\frac{8}{3}$ ✓

b. $-\frac{8}{3}$

c. $\frac{11}{3}$

d. 2

$\int_1^\infty 8x^{-4}\,dx = \lim_{b\to\infty}\left[\frac{8x^{-3}}{-3}\right]_1^b = \lim_{b\to\infty}\left(\frac{-8}{(3)b^3}+\frac{8}{3}\right) = 0+\frac{8}{3} = \frac{8}{3}$. Converges.

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34 [Intermediate] • Target: Improper Integrals with Infinite Limits (Type 1)

Evaluate $\int_1^\infty \frac{3}{x^4}\,dx$, or show it diverges.

a. $\frac{3}{4}$

b. 1 ✓

c. 2

d. -1

$\int_1^\infty 3x^{-4}\,dx = \lim_{b\to\infty}\left[\frac{3x^{-3}}{-3}\right]_1^b = \lim_{b\to\infty}\left(\frac{-3}{(3)b^3}+\frac{3}{3}\right) = 0+\frac{3}{3} = 1$. Converges.

35 [Intermediate] • Target: Improper Integrals with Discontinuous Integrands (Type 2)

Evaluate $\int_0^1 \frac{2}{x^{\frac{1}{4}}}\,dx$, or show it diverges (the integrand is undefined at $x=0$).

a. $\frac{8}{3}$ ✓

b. $\frac{11}{3}$

c. 8

d. $-\frac{8}{3}$

Since $p=\frac{1}{4}<1$, this converges (by the p-test at a discontinuity). $\int_0^1 2x^{-\frac{1}{4}}\,dx = \lim_{a\to 0^+}\left[\frac{2x^{\frac{3}{4}}}{\frac{3}{4}}\right]_a^1 = \frac{2}{\frac{3}{4}}-0 = \frac{8}{3}$.

36 [Intermediate] • Target: Partial fraction decomposition: distinct linear factors

Evaluate $\int_0^1 \frac{6}{x^{\frac{1}{5}}}\,dx$, or show it diverges (the integrand is undefined at $x=0$).

a. $\frac{15}{2}$ ✓

b. 30

c. $-\frac{15}{2}$

d. $\frac{17}{2}$

Since $p=\frac{1}{5}<1$, this converges (by the p-test at a discontinuity). $\int_0^1 6x^{-\frac{1}{5}}\,dx = \lim_{a\to 0^+}\left[\frac{6x^{\frac{4}{5}}}{\frac{4}{5}}\right]_a^1 = \frac{6}{\frac{4}{5}}-0 = \frac{15}{2}$.

37 [Intermediate] • Target: Partial fractions with a repeated linear factor

For $\frac{5x-9}{(x-4)^2} = \frac{A}{x-4}+\frac{B}{(x-4)^2}$, find B.

a. 11 ✓

b. -4

c. 13

d. -11

Multiplying both sides by $(x-4)^2$: $5x-9 = A(x-4)+B$. Setting $x=4$: $B = 5(4)-9 = 11$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Partial fractions with an irreducible quadratic factor • Cross-Lesson: 6.01 Power Series and Interval of Convergence

For $\frac{5x-3}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, find B.

a. -2

b. 4

c. 6

d. 2 ✓

Multiplying both sides by $(x-1)^2$: $5x-3 = A(x-1)+B$. Setting $x=1$: $B = 5(1)-3 = 2$.

39 [Challenging] • Target: Improper integrals: infinite limits • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Use a direct comparison to determine whether $\int_1^\infty \frac{2+\sin x}{x^2}\,dx$ converges or diverges.

a. Cannot be determined by comparison, since $\sin x$ oscillates.

b. Diverges, since $\frac{2+\sin x}{x^2} \geq \frac{1}{x^2}$ for $x \geq 1$, and $\int_1^\infty \frac{1}{x^2}\,dx$ diverges since $p = 2 \leq 1$.

c. **Converges, since $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$ for $x \geq 1$, and $\int_1^\infty \frac{3}{x^2}\,dx$ converges since $p = 2 > 1$.** ✓

d. Converges only because $a = 2$ happens to be even.

Since $-1 \leq \sin x \leq 1$: Converges, since $\frac{2+\sin x}{x^2} \leq \frac{3}{x^2}$ for $x \geq 1$, and $\int_1^\infty \frac{3}{x^2}\,dx$ converges since $p = 2 > 1$.

40 [Challenging] • Target: Improper integrals: discontinuous integrands • Cross-Lesson: 1.01 Integration by Parts

Use a direct comparison to determine whether $\int_1^\infty \frac{5+\sin x}{x}\,dx$ converges or diverges.

a. Diverges only because $a = 5$ happens to be even.

b. **Diverges, since $\frac{5+\sin x}{x} \geq \frac{4}{x}$ for $x \geq 1$, and $\int_1^\infty \frac{4}{x}\,dx$ diverges since $p = 1 \leq 1$.** ✓

c. Cannot be determined by comparison, since $\sin x$ oscillates.

d. Converges, since $\frac{5+\sin x}{x} \leq \frac{6}{x}$ for $x \geq 1$, and $\int_1^\infty \frac{6}{x}\,dx$ converges since $p = 1 > 1$.

Since $-1 \leq \sin x \leq 1$: Diverges, since $\frac{5+\sin x}{x} \geq \frac{4}{x}$ for $x \geq 1$, and $\int_1^\infty \frac{4}{x}\,dx$ diverges since $p = 1 \leq 1$.

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41 [Challenging] • Target: Linear Partial Fractions • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Evaluate $\int \frac{1}{x^2-4}dx$.

a. $\frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$ ✓

b. $\frac{1}{2}\ln\left|\frac{x-2}{x+2}\right| + C$

c. $\frac{1}{4}\ln|x^2-4| + C$

d. $\frac{1}{2}\arctan\left(\frac{x}{2}\right) + C$

Decomposition: $\frac{1}{(x-2)(x+2)} = \frac{1/4}{x-2} - \frac{1/4}{x+2}$. Integrating gives $\frac{1}{4}\ln|x-2| - \frac{1}{4}\ln|x+2| + C = \frac{1}{4}\ln\left|\frac{x-2}{x+2}\right| + C$.

42 [Challenging] • Target: Linear Partial Fractions • Cross-Lesson: 3.01 Parametric Equations

When must long division be performed BEFORE applying partial fractions?

a. **When the degree of the numerator is greater than or equal to the degree of the denominator** ✓

b. When the denominator has repeated roots

c. When the numerator is a constant

d. Always

Partial fractions applies only to proper rational functions. If $\deg(P) \geq \deg(Q)$, polynomial long division must be performed first.

43 [Challenging] • Target: Repeated Linear & Irreducible Factors • Cross-Lesson: 3.02 Vector-Valued Functions

Evaluate $\int_2^3 \frac{1}{x(x-1)}dx$.

a. $\ln\left(\frac{4}{3}\right)$ ✓

b. $\ln 2$

c. $\ln 3$

d. $\frac{1}{6}$

$\frac{1}{x(x-1)} = \frac{1}{x-1} - \frac{1}{x}$. $\left[\ln\left|\frac{x-1}{x}\right|\right]_2^3 = \ln\left(\frac{2}{3}\right) - \ln\left(\frac{1}{2}\right) = \ln\left(\frac{4}{3}\right)$.

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44 [Challenging] • Target: Improper Integrals with Infinite Limits (Type 1) • Cross-Lesson: 4.01 Polar Curves and Derivatives

For what values of p does the improper integral $\int_1^\infty \frac{1}{x^p}dx$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. All real p

By the p-integral test, $\int_1^\infty \frac{1}{x^p}dx$ converges if and only if $p > 1$ (to $\frac{1}{p-1}$), and diverges for $p \leq 1$.

45 [Challenging] • Target: Improper Integrals with Discontinuous Integrands (Type 2) • Cross-Lesson: 4.02 Area in Polar Coordinates

Evaluate the improper integral $\int_0^1 \frac{1}{\sqrt{x}}dx$.

a. 2 ✓

b. 1

c. ∞ (diverges)

d. $\frac{1}{2}$

The integrand has an infinite discontinuity at $x = 0$: $\lim_{a \to 0^+} [2\sqrt{x}]_a^1 = 2(1) - 0 = 2$.

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